

LESSON 5-4 • SESSION 2 • PRACTICE • FORM B

Composite Figures

Learning target – I can find the area of a composite figure by adding or subtracting the areas of simpler shapes. · 6.G.A.1

REMEMBER

- A **composite figure** is made of simpler shapes: rectangles, triangles, squares.
- **Add** the areas of pieces that are joined. **Subtract** the area of a piece that is cut out.
- Use each piece's own formula: rectangle $l \times w$, triangle $\frac{1}{2} \times b \times h$.

WORKED EXAMPLE

A 14 cm by 9 cm rectangle has a triangle (base 6 cm, height 4 cm) cut out.

Step 1 Whole rectangle: $14 \times 9 = 126 \text{ cm}^2$.

Step 2 Cut-out triangle: $\frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$.

Step 3 Removed, so subtract: $126 - 12$.

Answer **114 cm² left**

WORDS **composite figure** (*figura compuesta*) · **add** (*sumar*) · **subtract** (*restar*)

PRACTICE

show your work · label units

1. Decide whether to add or subtract. Then find each area.

a. 36 cm² rectangle + 10 cm² triangle ____ cm² b. 81 in² square, 25 in² cut out ____ in²

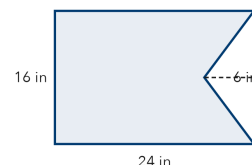
c. 8 ft square, 2 ft square cut out ____ ft² d. 12 m × 5 m, 3 m square cut out ____ m²

2. **Swallowtail Flag.** A flag starts as a rectangle 24 in by 16 in. A triangle notch 6 in deep is cut out of the right edge.

a. Area of the whole rectangle ____ in²

b. Area of the notch ____ in²

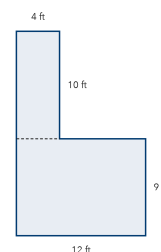
c. Area of the flag ____ in²



3. **L-Shaped Room.** An L-shaped floor splits into a 12 ft by 9 ft rectangle and a 4 ft by 10 ft rectangle. Find the floor area.

Show your work.

Answer _____ ft²



4. **Wall and Window.** A wall is 18 ft long and 9 ft tall. A square window 4 ft on each side is not painted. Find the painted area.

Hint The window is taken away.

Show your work.

Answer _____ ft²

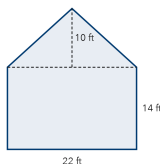
5. **Plywood Sheet.** A plywood sheet is 20 ft by 8 ft. A 5 ft by 3 ft opening is cut out. How much plywood is left?

Show your work.

Answer _____ ft²

6. **House Front.** A house front is a rectangle with a triangle roof. Find the total area.

Hint Use each shape's own formula, then add.



Show your work.

Answer _____ ft^2

7. **Nico's Poster.** A poster is 30 in by 24 in. An 8 in square is cut from one corner. Nico says 720 in^2 is left. Is Nico right?

Hint Did the corner square get taken away?

Show your work.

Area left _____

8. **Playground Plan.** A playground is 28 m by 18 m. Rubber covers all of it except a triangle sandbox (base 10 m, height 8 m). Rubber costs \$12 per m^2 .

- a. Area of the whole rectangle _____ m^2
 b. Area covered with rubber _____ m^2
 c. Cost of the rubber _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Lila's Work.** A wall is 17 ft by 10 ft with a 4 ft by 4 ft vent that is not painted. Lila wrote: "Painted area = $170 + 16 = 186 \text{ ft}^2$." Explain the mistake and fix it.

TEST PRACTICE

choose the best answer

10. A 64 in^2 square with a 9 in^2 square cut out has an area of –
A 55 in^2 **B** 73 in^2 **C** 61 in^2 **D** 576 in^2
11. A 50 cm^2 rectangle joined to a 15 cm^2 triangle has a total area of –
A 35 cm^2 **B** 50 cm^2 **C** 65 cm^2 **D** 750 cm^2

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

656 N	432 Q	146 M	784 Z	5,568 D	464 E	336 C	418 I	148 O	145 B
#2c	#3	#4	#5	#6	#7	#8b	#8c		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____